PROJECT REPORT

Title	Script-Controlled ACL – Restrict Record Access Based on Field Value
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1. INTRODUCTION

1.1 Project Overview

This project implements script-controlled Access Control Lists (ACLs) in ServiceNow to restrict access to student and institution records based on user roles and the Branch field.

1.2 Purpose

The purpose is to protect records from unauthorized viewing, creation, modification, and deletion while allowing administrators full access.

2. IDEATION PHASE

2.1 Problem Statement

Unauthorized users may access student and institution records. The project provides role-based and field-based access control for these records.

2.2 Brainstorming

- Create users and different roles.
- Create an Institution Details table.
- Use ACLs to control access.
- Restrict EEE records using the Branch field.
- Use separate ACLs for Read, Create, Write, and Delete.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey map

Login → Access Records → Check ACL → Perform permitted operation → Access Granted or Denied.

3.2 Solution Requirement

- Create users and roles such as bb1, bb2, bb3, and bb4.
- Create the Institution Details table and student records.
- Control Read, Create, Write, and Delete operations.
- Allow administrators full access.

3.3 Data Flow Diagram

User → ServiceNow → ACL Check → Institution Details Table → Access Granted / Denied.

3.4 Technology Stack

ServiceNow, ServiceNow ACL, JavaScript, ServiceNow Tables, and ServiceNow Roles.

4. PROJECT DESIGN

4.1 Problem Solution Fit

The solution combines user roles, ACLs, and the Branch field to prevent unauthorized access to records.

4.2 Proposed Solution

A Script-Controlled ACL system is implemented in ServiceNow. Read, Create, Write, and Delete permissions are controlled using different roles.

4.3 Solution Architecture

User → Authentication → User/Role Management → ACL → Role and Condition Check → Institution Details → Access Decision.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project was planned through requirement identification, user and role creation, table creation, record creation, ACL configuration, testing, and verification.

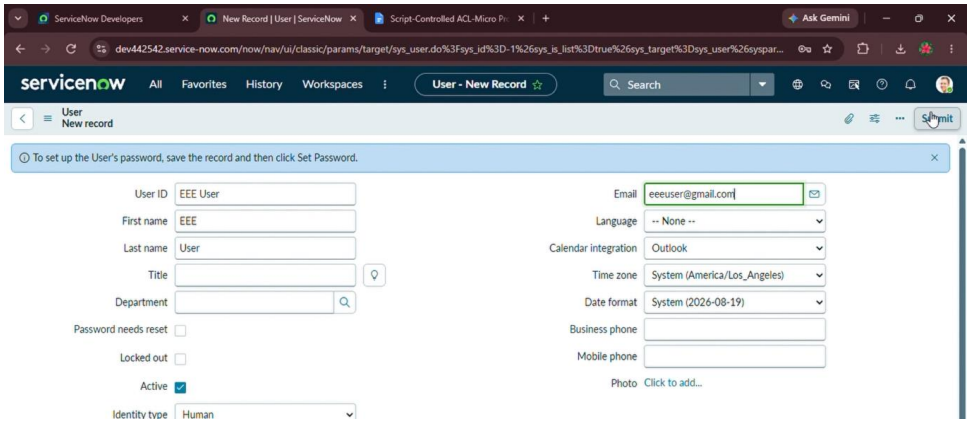
6. FUNCTIONAL AND PERFORMANCE TESTING

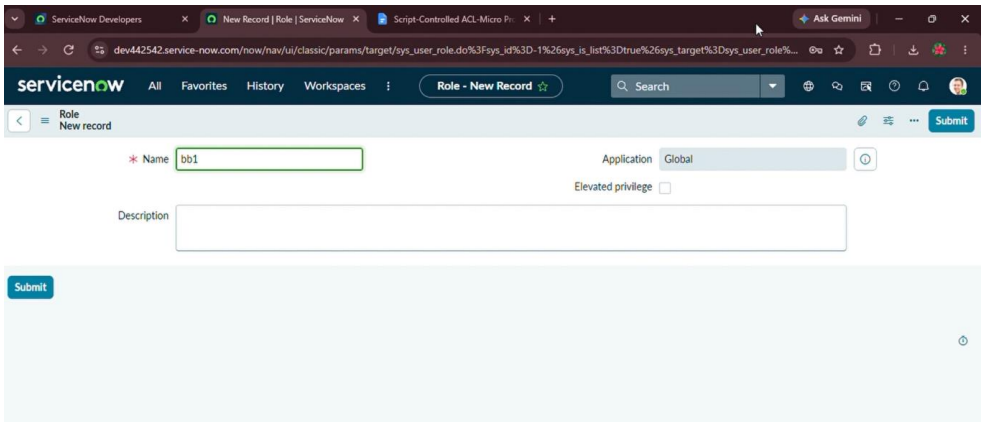
6.1 Performance Testing

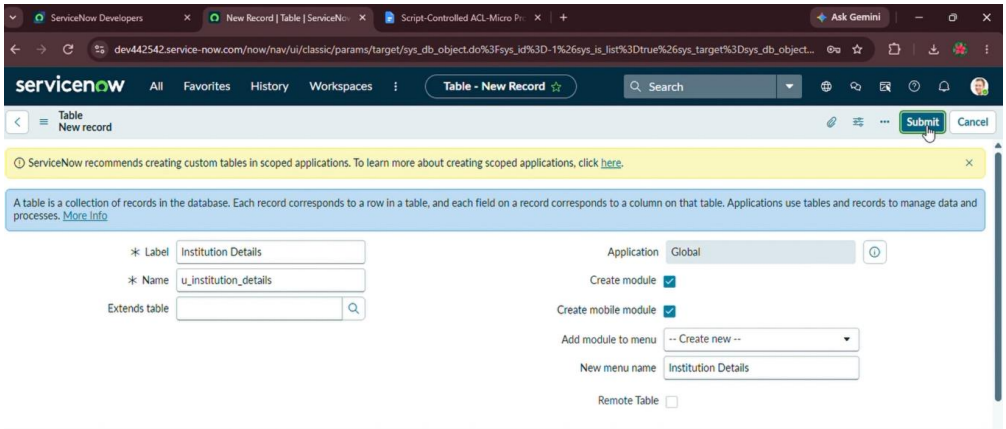
The ACLs were tested with users having different roles. Read, Create, Write, and Delete operations were checked to verify that access was provided or denied correctly.

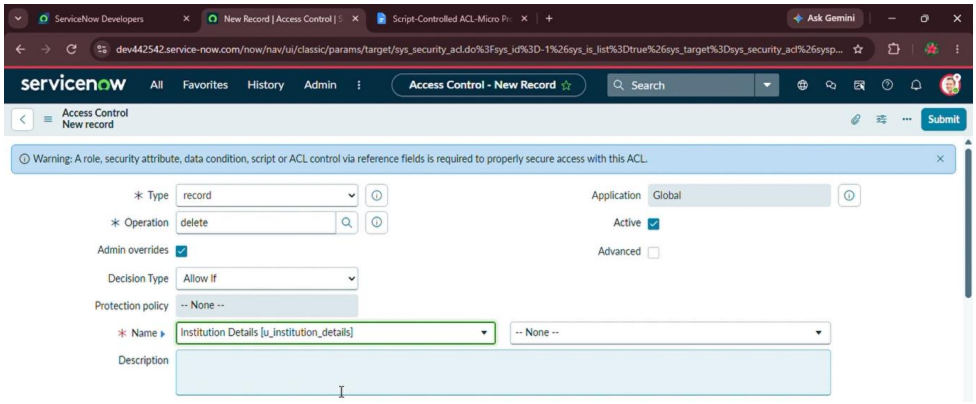
7. RESULTS

7.1 Output Screenshots









Screenshots of user creation, roles, Institution Details table, ACL configurations, and access verification can be included here.

8. ADVANTAGES & DISADVANTAGES

- Provides secure record-level access.
- Supports different user permissions.
- Controls Read, Create, Write, and Delete operations.
- ACL configuration can become complex when many roles are added.

9. CONCLUSION

The project successfully demonstrates how ServiceNow ACLs can control access to student and institution records using roles and conditions. It helps prevent unauthorized viewing and modification of records.

10. FUTURE SCOPE

The system can be extended with more roles, branches, access conditions, and additional ServiceNow security rules.

11. APPENDIX

Source Code (if any): (function () {

```
// Allow admin users full access
if (gs.hasRole('admin')) {
    return true;
}
// Allow only EEE branch users to see EEE records
if (gs.hasRole('bb1')){
    return true;
}
```

```
    // Deny access for all others
    return false;
})();
```